

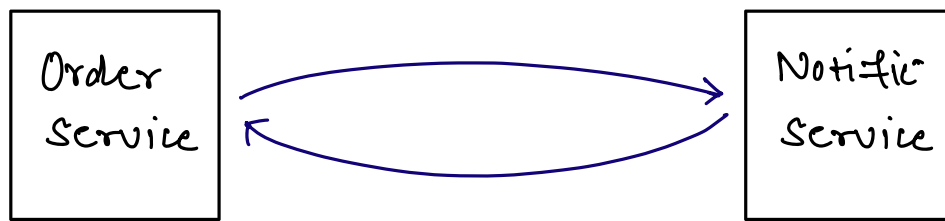
Agenda.

→ Message Queues.

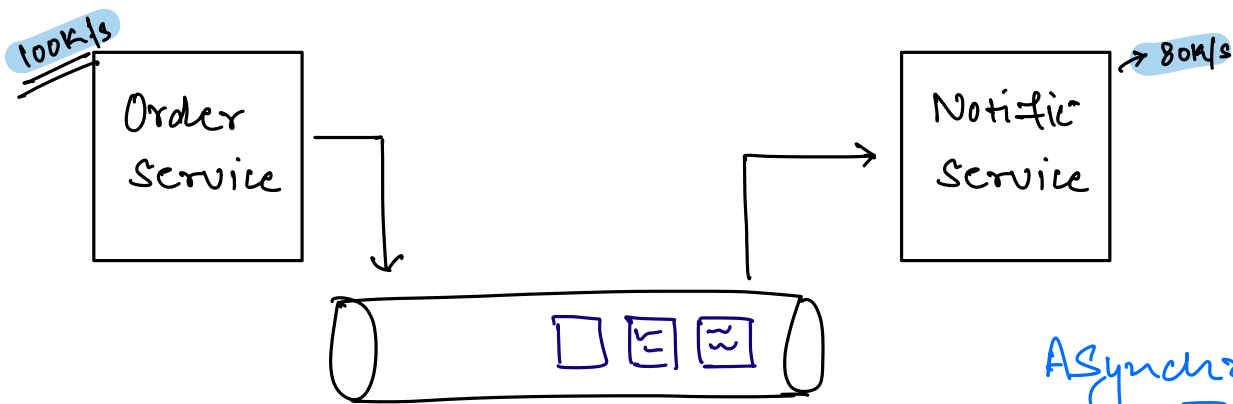
↳ Kafka.

→ Zookeeper.

#



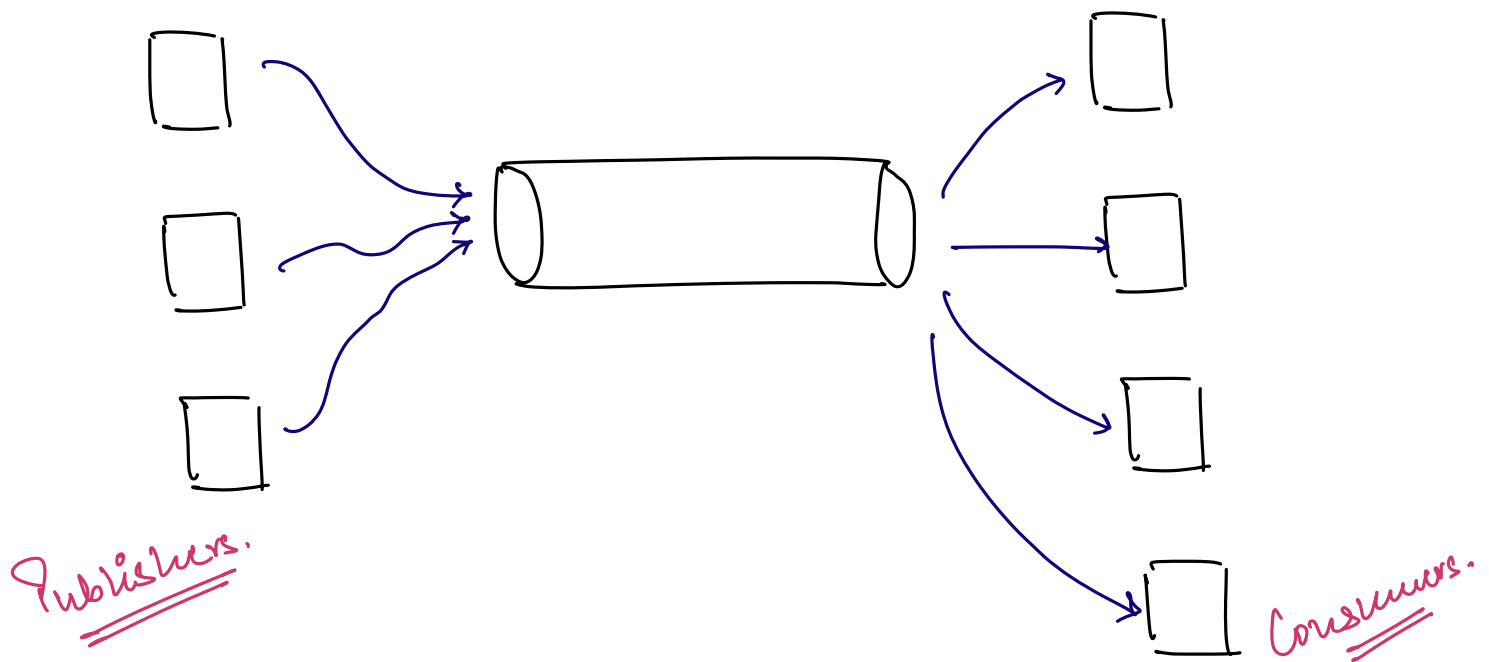
Synchronous.



Asynchronous.

⇒ Publisher / Subscriber (Pub/Sub)

↳ Producer / Consumer.



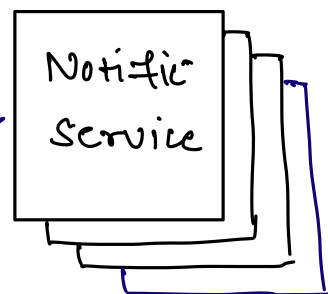
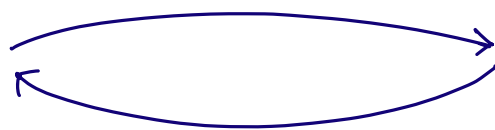
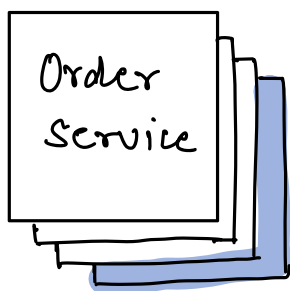
How.
→ Pub/Sub (vs)
Observer.

Advantages of Pub-Sub.

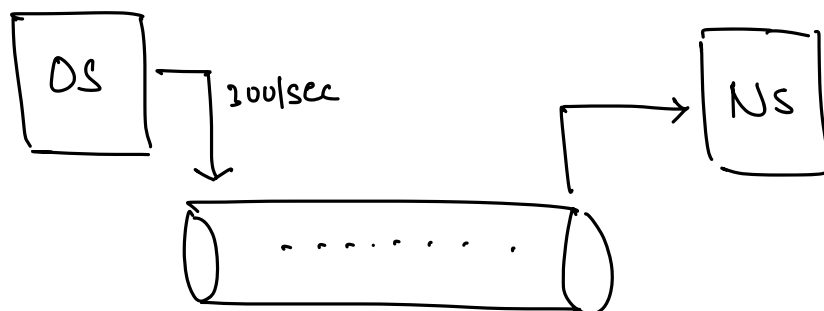
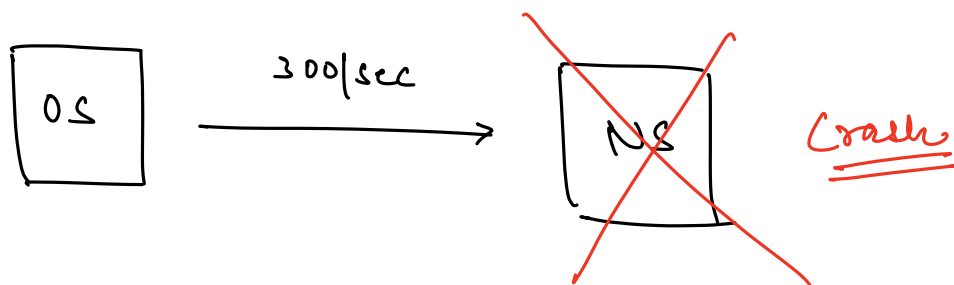
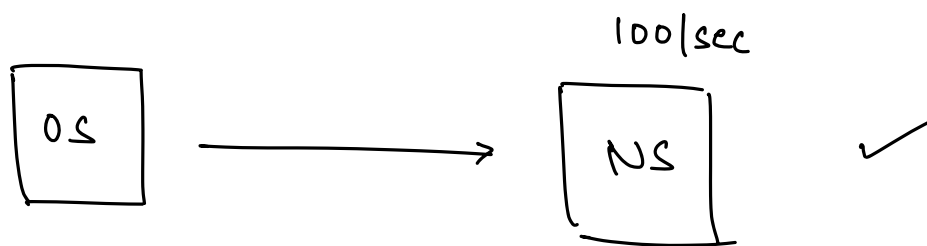
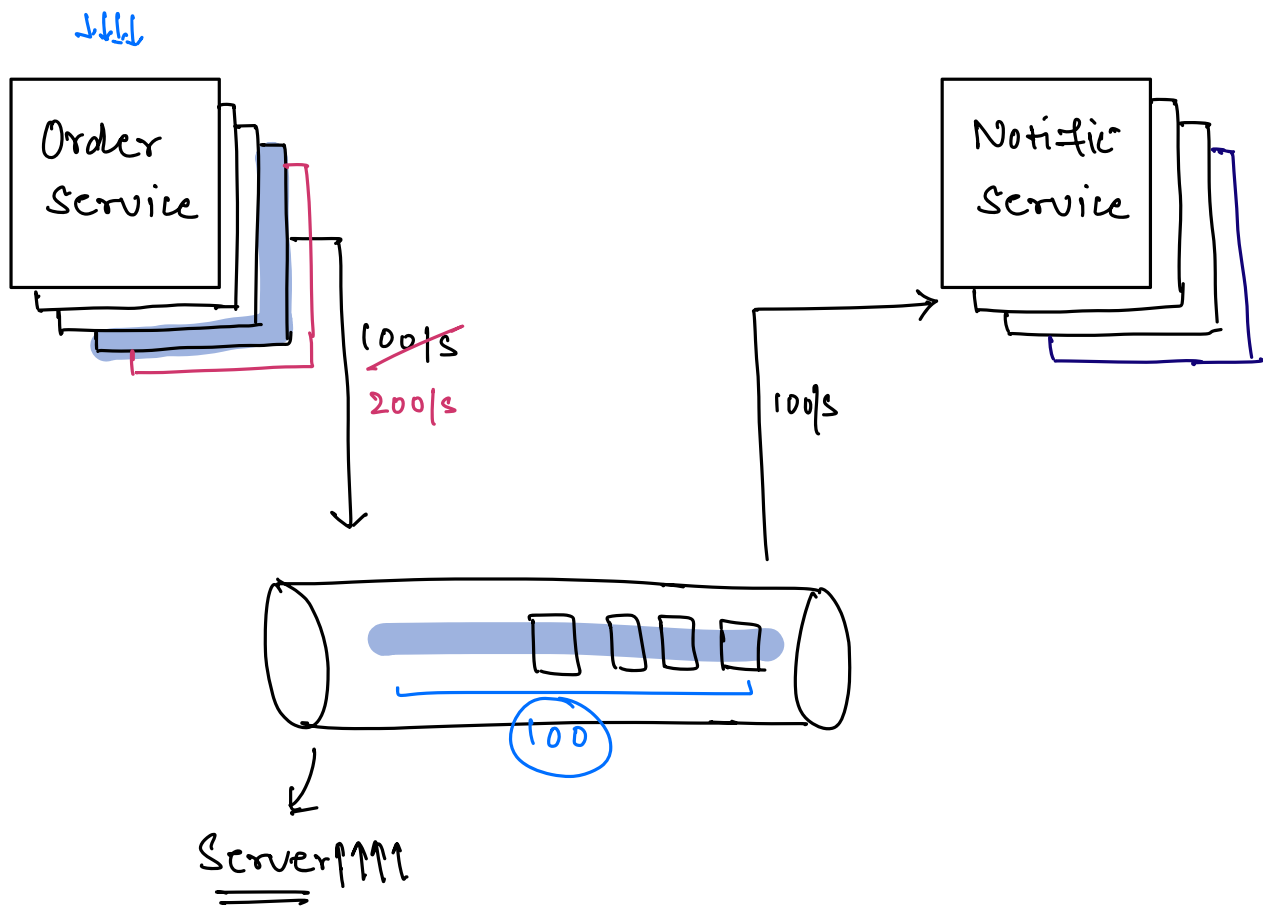
1. Decoupling.

↳ Scale the services individually.

100 → 200

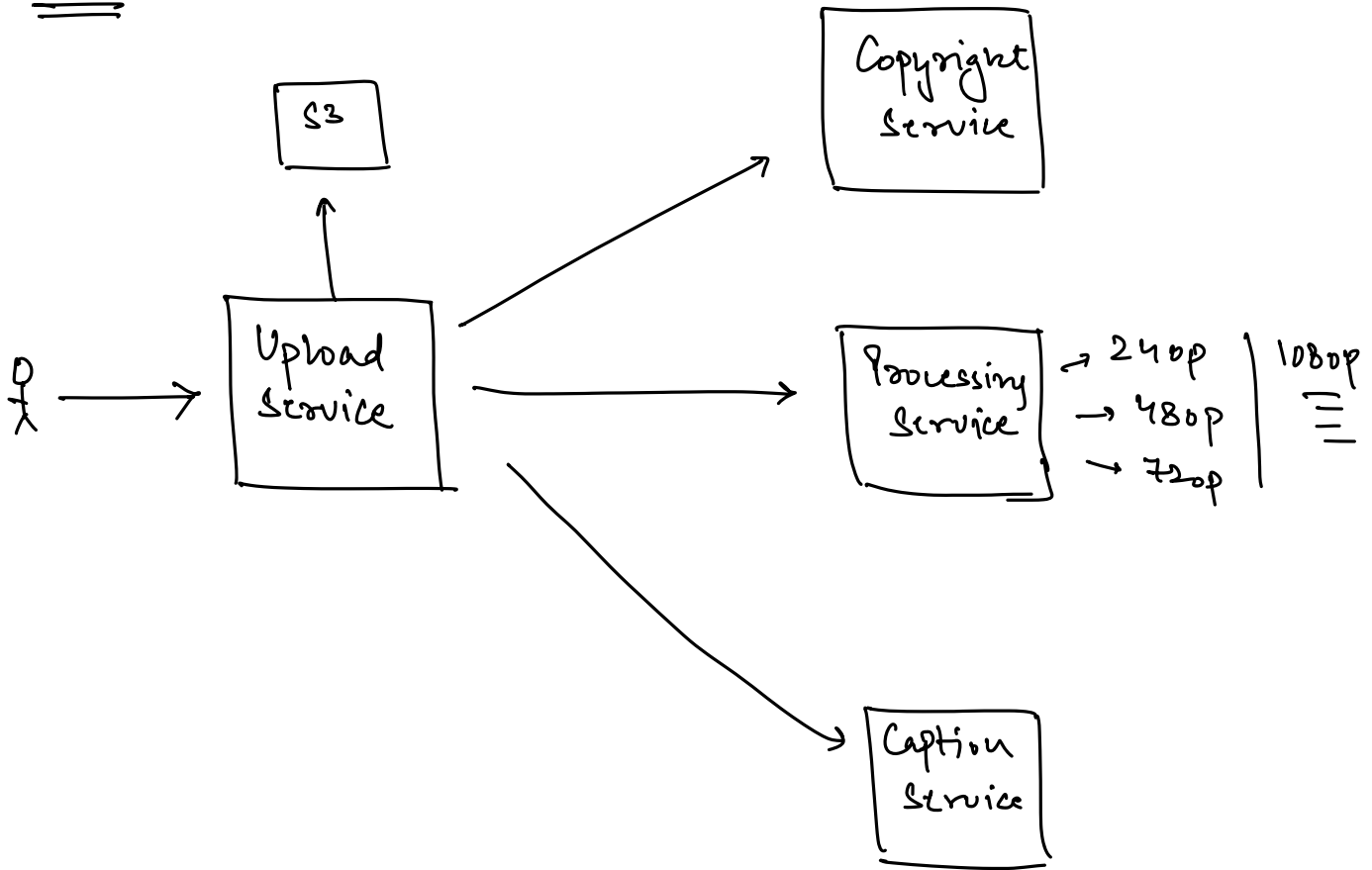


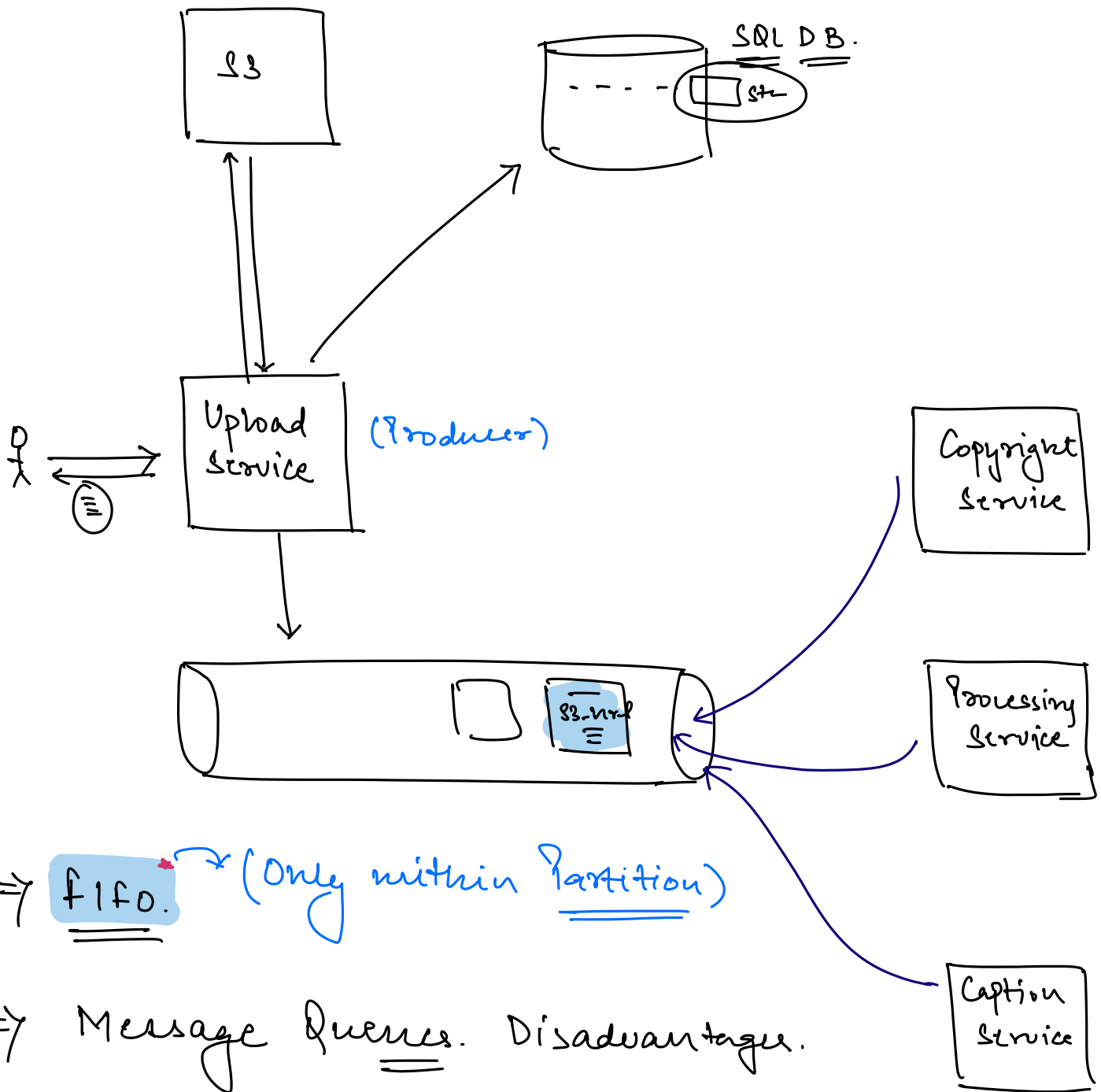
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⇒ Persistent Queues.

Youtube.

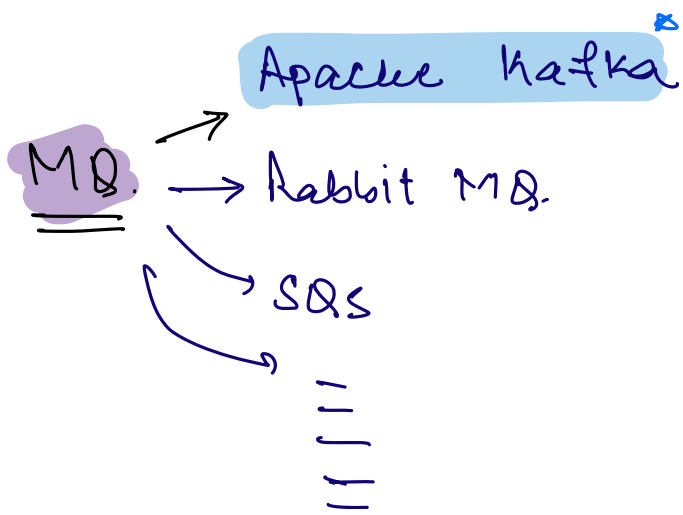




⇒ fifo. (Only within Partition)

⇒ Message Queues. Disadvantages.

- Only Async.
- latency $\uparrow\uparrow$ (Compare to Sync)
- N/w overhead.



Apache Kafka.
→ LinkedIn.

→ High throughput → 10M msg/day.

→ Ultra low latency.

→ Horizontally scalable.

→ Fault Tolerant.

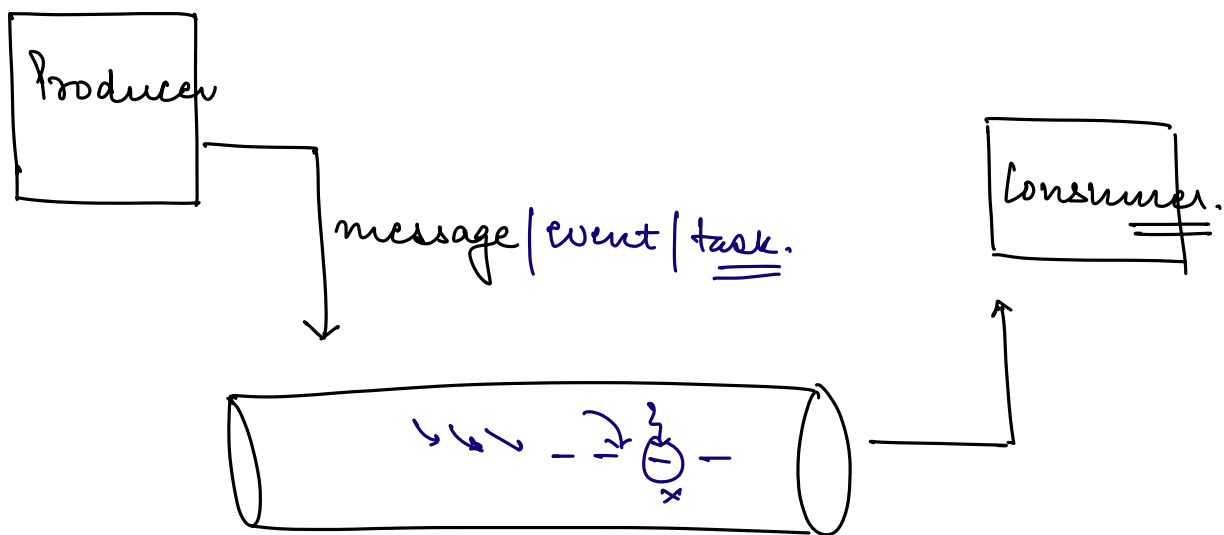
→ High Availability.

→ Persistent.

→ Message Ordering*

↳ fifo.

→ Kafka provides strict message Ordering (only within partition)



⇒ Kafka is append only.

⇒ msg → few KBs.

topic: _____

Value: 1
 |||||
3

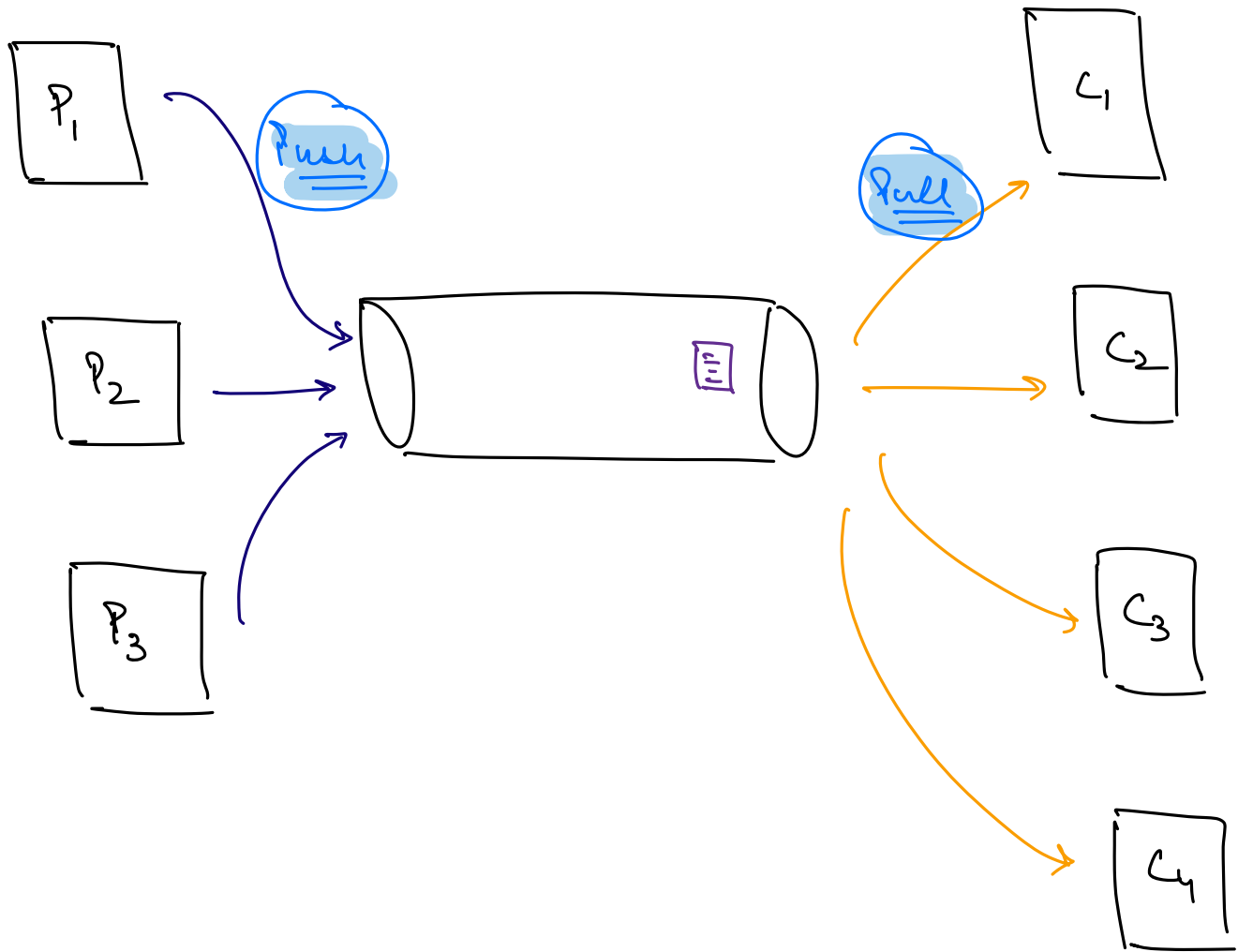
Partition ← key:

timestamp: _____

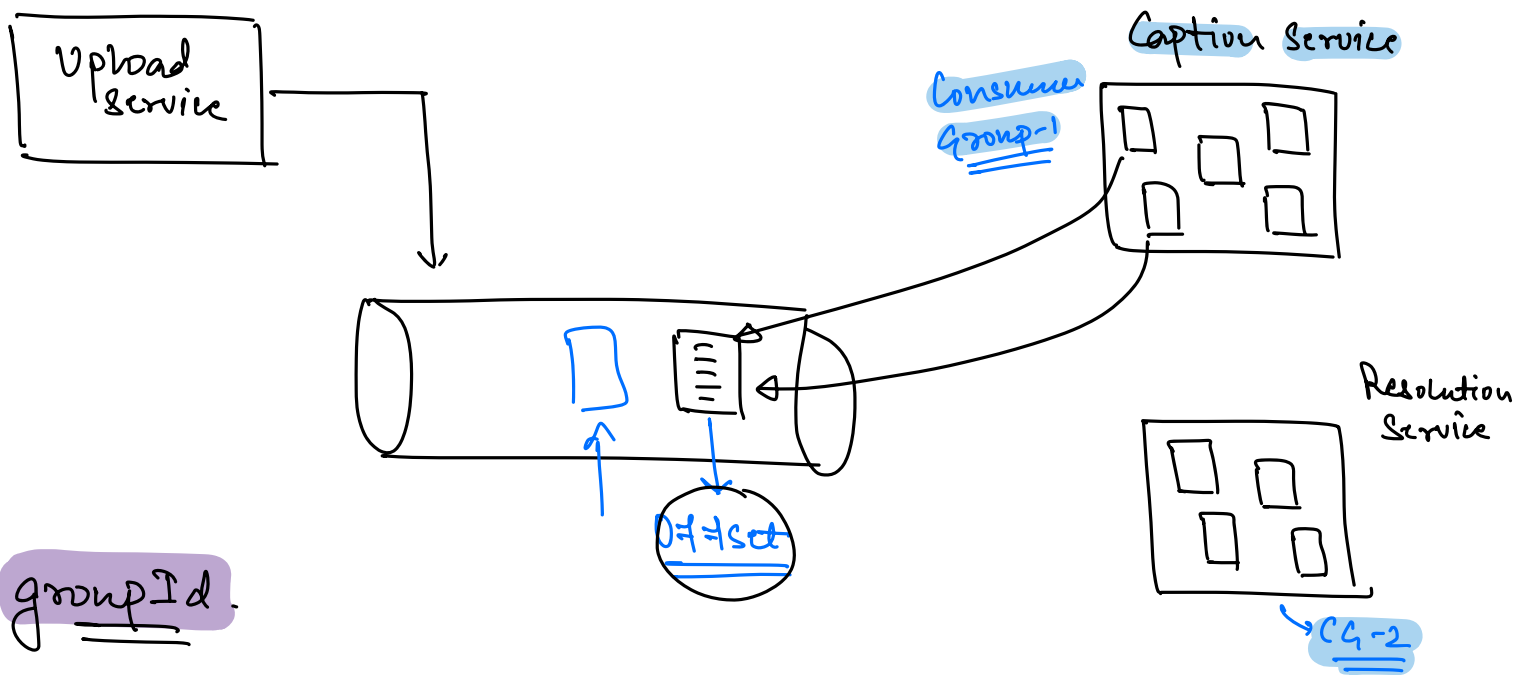
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⇒ Kafka : Push & Pull based.

└→ Push on producer side
 └→ Pull on consumer side.

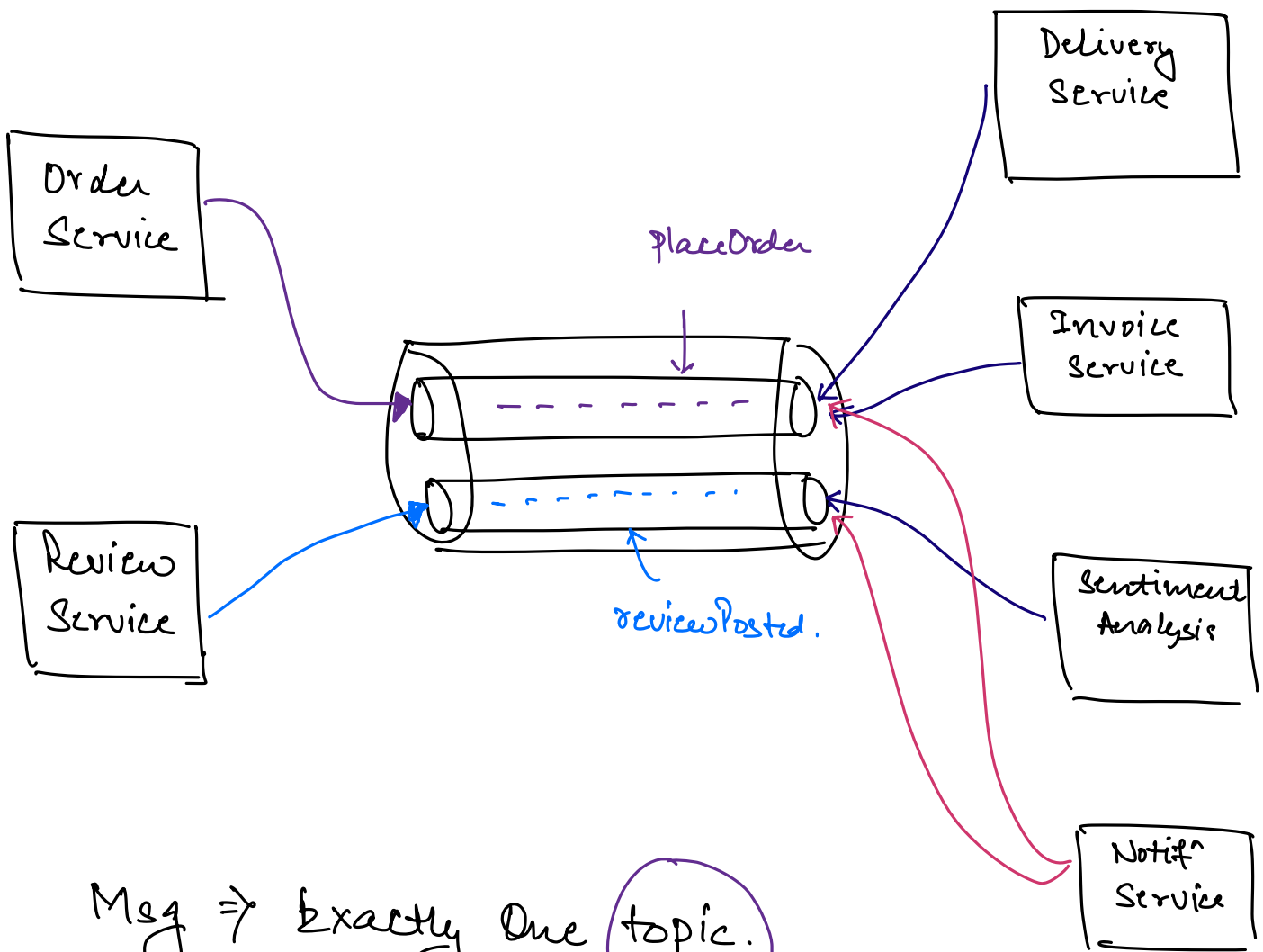


#



⇒ groupId.

Topic.



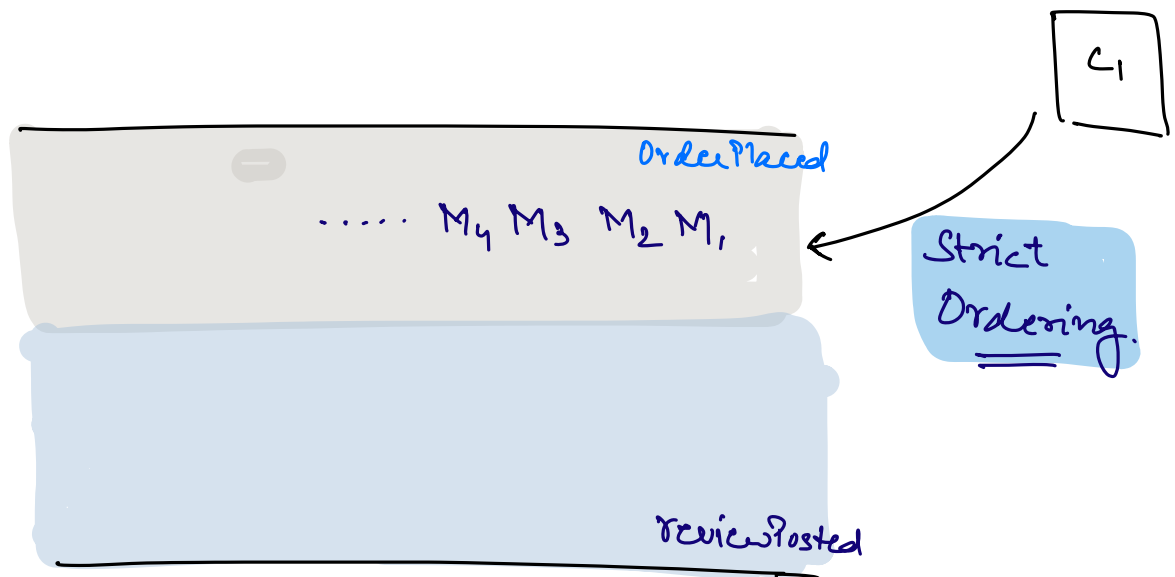
Msg $\Rightarrow$ Exactly One topic.
Subscribers

→ Consumer Group can decide on which topic they want to read from.

→ Consumer Group can read the messages from any no. of topics.

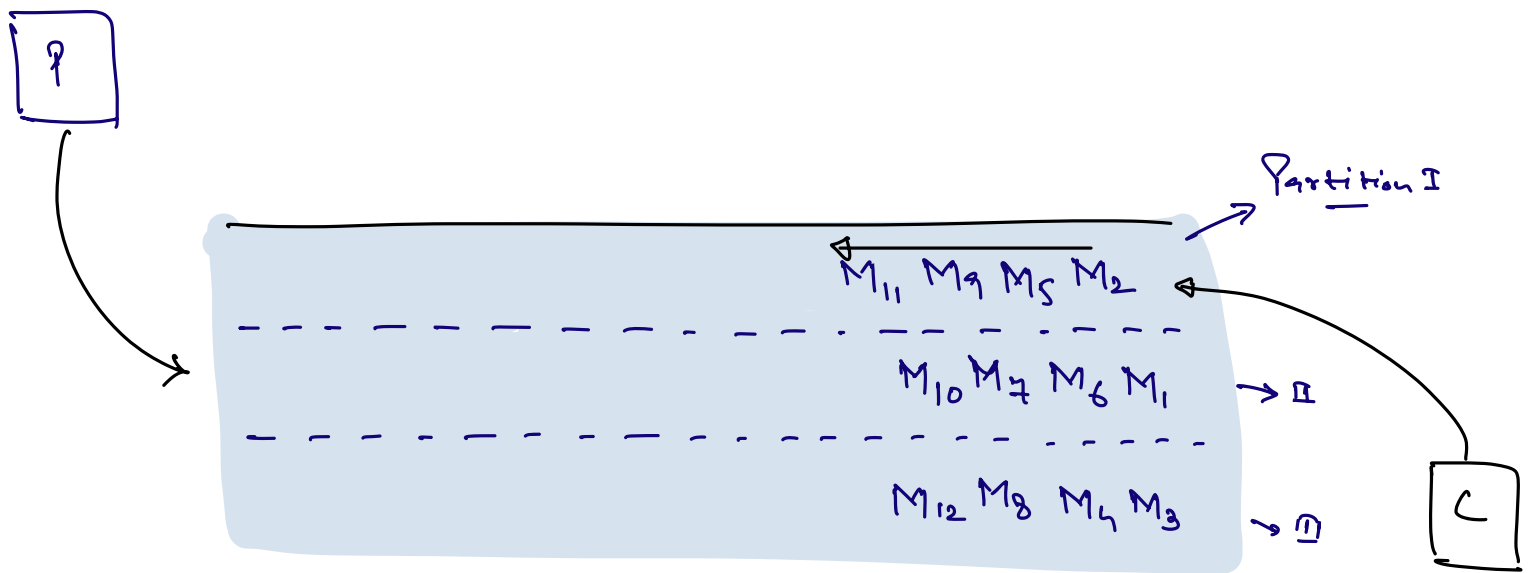
→ When producer inserts a msg into kafka, it must provide topic.

Topics in kafka are multi-producer & multi-consumer, a topic can have 0, one or more producers that writes to it & 0, one or more consumers that consumes from it.



⇒ A single server might not be enough to store all the messages of 1 topic.

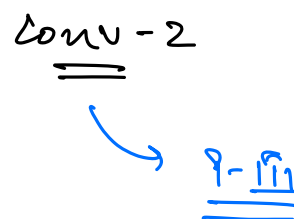
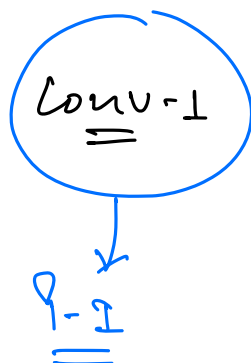
⇒ Shard the Topic : Partition



⇒ Kafka doesn't guarantee ordering across partitions.

⇒ Kafka only guarantees ordering within partition.

Chat App



Uber Driver

msg
↳ t

topic: _____

Value: t
=====

Partition ← key (Optional field)

timestamp: _____

=====

————— * —————